

Inés Alejandro Cruz Guerrero

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Current Position

Postdoctoral Fellow

2026 - Present

Department of Neurological Surgery, Biological Science Division
University of Chicago

Contact Information

Office address: Gilman Smith Building, W350, 3rd Floor, 5841 S. Maryland Ave., Chicago, Illinois, 60615, U.S.A..
E-mail: alejandro.cruz@bsd.uchicago.edu

Education

Autonomous University of San Luis Potosi (UASLP)

S.L.P., Mexico

PhD's Degree in Engineering Science

Sep. 2019 - 2023

Autonomous University of San Luis Potosi (UASLP)

S.L.P., Mexico

Master's Degree in Electronics Engineering

Sep. 2017 - Aug. 2019

Autonomous University of San Luis Potosi (UASLP)

S.L.P., Mexico

Bachelor's Degree in Biomedical Engineering

Sep. 2012 - Jun. 2017

Technical Skills

Programming

Python, C++, C, C#, VHDL, MATLAB

AI / ML

PyTorch, TensorFlow, Scikit-learn, NumPy, SciPy

Engineering Software

OpenCV, VTK, MIPAV, LabVIEW, Simulink, Multisim, PCBWizard, Easy-PC, SolidWorks, BioPac

Productivity

Microsoft Office, LibreOffice, L^AT_EX, Mendeley, Zotero, OBS Studio

Languages

Spanish (Native), English (B2)

Publications

- [1] **Cruz-Guerrero, I. A.**, Nagel, J. & Porras, A. R. (2026). "Unsupervised adaptive sampling graph autoencoder for 3D surface encoding and mesh representation transfer", in *Medical Image Analysis*.
- [2] Palmer, S. K., Elkhill, C., **Cruz-Guerrero, I. A.**, Gomez, D. A., Linguraru, M. G., Wilkinson, C., Khechoyan, D., Nguyen, P., French, B., & Porras, A. R. (2026). "Data-driven personalized surgical treatment selection for metopic and sagittal craniosynostosis", in *Journal of Neurosurgery: Pediatrics*, 1–10.
- [3] Liu, J., Xing, F., Elkhill, C., Linguraru, M. G., Miles, R. C., **Cruz-Guerrero, I. A.**, & Porras, A. R. (2025). "Population-driven synthesis of personalized cranial development from cross-sectional pediatric CT images", in *IEEE Transactions on Biomedical Engineering*, 72(9), 2732–2741.
- [4] Palmer, S. K., Elkhill, C., **Cruz-Guerrero, I. A.**, Gomez, D. A., Wilkinson, C., Khechoyan, D., French, B., Porras, A. R., & Nguyen, P. (2025). "Early outcomes of endoscopic spring-assisted repair of metopic craniosynostosis", in *The Journal of Craniofacial Surgery*, 36(5), 1480.
- [5] Mendoza-Chavarria, J. N., **Cruz-Guerrero, I. A.**, Gutiérrez-Navarro, O., León, R., Ortega, S., Fabelo, H., Callicó, G. M., & Campos-Delgado, D. U. (2024). "Blind non-linear spectral unmixing with spatial coherence for hyper and multispectral images", in *Journal of the Franklin Institute*, 361(18), 107282.
- [6] Görg, C., Elkhill, C., Chaij, J., Royalty, K., Nguyen, P. N., French, B., **Cruz-Guerrero, I. A.**, & Porras, A. R. (2024). "SHAPE: a visual computing pipeline for interactive landmarking of 3D photograms and patient reporting for assessing craniosynostosis", in *Computers & Graphics*, 125, 104056.
- [7] **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Mejía-Rodríguez, A. R., León, R., Ortega, S., Fabelo, H., & Callicó, G. M. (2024). "Hybrid brain tumor classification of histopathology hyperspectral images by linear unmixing and an ensemble of deep neural networks", in *Health-care Technology Letters*, 11(4), 240–251.
- [8] **Cruz-Guerrero, I. A.**, Mejía-Rodríguez, A. R., Ortega, S., Fabelo, H., Callicó, G. M., Jo, J. A., & Campos-Delgado, D. U. (2023). "Multi and hyperspectral image unmixing with spatial coherence by extended blind end-member and abundance extraction", in *Journal of the Franklin Institute*, 360(15), 11165–11196.
- [9] León, R., Fabelo, H., Ortega, S., **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Szolna, A., & Callicó, G. M. (2023). "Hyperspectral imaging benchmark based on machine learning for intraoperative brain tumour detection", in *npj Precision Oncology*, 7(1), 119.
- [10] Campos-Delgado, D. U., **Cruz-Guerrero, I. A.**, Mendoza-Chavarria, J. N., Mejía-Rodríguez, A. R., Ortega, S., Fabelo, H., & Callicó, G. M. (2022). "Nonlinear extended blind end-member and abundance extraction for hyperspectral images", in *Signal Processing*, 201, 108718.
- [11] **Cruz-Guerrero, I. A.**, León, R., Campos-Delgado, D. U., Ortega, S., Fabelo, H., & Callicó, G. M. (2020). "Classification of hyperspectral in vivo brain tissue based on linear unmixing", in *Applied Sciences*, 10(16), 5686.

Publications of Book Chapters

- **Cruz-Guerrero, I. A.**, León, R., Mejía-Rodríguez, A. R., Campos-Delgado, D. U., Fabelo, H., Ortega, S., & Callicó, G. M. (2023). “Hyperspectral imaging for cancer applications”, in *Diagnosis and Treatment of Cancer Using Thermal Therapies*, pp. 81–101. CRC Press.
- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Mejía-Rodríguez, A. R., Fabelo, H., Ortega, S., & Callicó, G. M. (2022). “A hybrid approach to the hyperspectral classification of in vivo brain tissue: linear unmixing with spatial coherence and machine learning”, in *Artificial Intelligence in Cancer Diagnosis and Prognosis: Brain and Prostate Cancer*, Vol. 3, Ch. 9, pp. 9–1–9–22. IOP Publishing.

Publications in International Conferences

- Elkhill, C., **Cruz-Guerrero, I. A.**, Khechoyan, D., Nguyen, P. N., French, B., & Porras, A. R. (2026). “A temporally aware transformer model for predicting personalized head growth and surgical outcomes in craniosynostosis”, in *2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*, pp. 1–4. IEEE.
- Elkhill, C., **Cruz-Guerrero, I. A.**, Linguraru, M. G., French, B., & Porras, A. R. (2025). “A generative model with domain adversarial training and independent conditional discriminators to predict the development of craniosynostosis”, in *2025 21st Symposium on Medical Information Processing and Analysis (SIPAIM)*, pp. 1–4. IEEE.
- Elkhill, C., **Cruz-Guerrero, I. A.**, Liu, J., Linguraru, M. G., Alexander, A., French, B., & Porras, A. R. (2024). “Personalized phenotype encoding and prediction of pathological head development from cross-sectional images”, in *2024 20th Symposium on Medical Information Processing and Analysis (SIPAIM)*, pp. 1–4. IEEE.
- **Cruz-Guerrero, I. A.**, Elkhill, C., Liu, J., Nguyen, P. N., French, B., & Porras, A. R. (2024). “Mesh registration via geometric feature homogenization and offset cross-attention: application to 3D photogrammetry”, in *International Workshop on Graphs in Biomedical Image Analysis (GRAIL 2024)*, pp. 96–105. Springer, Cham.
- Mendoza-Chavarria, J. N., **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Fabelo, H., Ortega, S., & Callicó, G. M. (2023). “Ensemble of artificial intelligence classifiers for in-vivo identification of glioblastoma tumours using hyperspectral images”, in *2023 1st IEEE Engineering in Medicine and Biology Conference in Latin America (IEEE EMBC R9)*. IEEE.
- **Cruz-Guerrero, I. A.**, Mendoza-Chavarria, J. N., Campos-Delgado, D. U., Fabelo, H., Ortega, S., & Callicó, G. M. (2023). “Classification of brain tissues in hyperspectral images using vision transformers”, in *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*. IEEE.
- **Cruz-Guerrero, I. A.**, León, R., Granados-Castro, L., Fabelo, H., Ortega, S., Campos-Delgado, D. U., & Callicó, G. M. (2022). “Reflectance calibration with normalization correction in hyperspectral imaging”, in *2022 25th Euromicro Conference on Digital System Design (DSD)*. IEEE.
- Mendoza-Chavarria, J. N., Zavala-Sánchez, E. R., Granados-Castro, L., **Cruz-Guerrero, I. A.**, Fabelo, H., Ortega, S., Callicó, G. M., & Campos-Delgado, D. U. (2022). “Glioblastoma classification in hyperspectral images by nonlinear unmixing”, in *2022 25th Euromicro Conference on Digital System Design (DSD)*. IEEE.
- Granados-Castro, L., Gutiérrez-Navarro, O., **Cruz-Guerrero, I. A.**, Mendoza-Chavarria, J. N., Zavala-Sánchez, E. R., & Campos-Delgado, D. U. (2022). “Estimation of deoxygenated and oxygenated hemoglobin by multispectral blind linear unmixing”, in *2022 25th Euromicro Conference on Digital System Design (DSD)*. IEEE.
- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., & Mejía-Rodríguez, A. R. (2021). “Extended blind end-member and abundance estimation with spatial total variation for hyperspectral imaging”, in *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, pp. 1957–1960. IEEE.
- Alvarez-Gutiérrez, M. T., Mejía-Rodríguez, A. R., **Cruz-Guerrero, I. A.**, & Arce-Santana, E. R. (2019). “3D kidney reconstruction from 2D ultrasound images”, in *Latin American Conference on Biomedical Engineering*, pp. 393–400. Springer, Cham.

Publications in National Conferences

- Martínez-Arboleyda, D. N., **Cruz-Guerrero, I. A.**, & Campos-Delgado, D. U. (2023). “Tumor tissue classification in hyperspectral histopathology images through individual and ensemble of machine learning algorithms”, in *XLVI Mexican Conference on Biomedical Engineering (CNIB 2023)*, pp. 100–111. Springer, Cham.
- **Cruz-Guerrero, I. A.**, Mendoza-Chavarria, J. N., & Campos-Delgado, D. U. (2022). “Glioblastoma classification in hyperspectral images by reflectance calibration with normalization correction and nonlinear unmixing”, in *XLV Mexican Conference on Biomedical Engineering (CNIB 2022)*, Vol. 86. Springer, Cham.
- Mendoza-Chavarria, J. N., **Cruz-Guerrero, I. A.**, Mejía-Rodríguez, A. R., & Campos-Delgado, D. U. (2021). “Blind unmixing algorithm based on the multi-linear mixture model”, in *XLIV Mexican Conference on Biomedical Engineering (CNIB 2021)*, 8(1), 106–109.
- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., Mejía-Rodríguez, A. R., Ortega, S., Fabelo, H., & Marrero Callicó, G. M. (2021). “Hybrid approach to histological specimen classification for brain tumor detection: linear unmixing and neural networks”, in *XLIV Mexican Conference on Biomedical Engineering (CNIB 2021)*, 8(1), 97–100.
- **Cruz-Guerrero, I. A.**, Campos-Delgado, D. U., & Mejía-Rodríguez, A. R. (2020). “Blind estimation of total variance endmembers and abundances for hyperspectral imaging for clinical use”, in *XLIII Mexican Conference on Biomedical Engineering (CNIB 2020)*, 7(1), 548–555.

Experience

Postdoctoral Researcher

Department of Biostatistics and Informatics, University of Colorado Anschutz Medical Campus

Aurora, CO

Oct. 2023 - Mar. 2026

Research Internship

Institute of Applied Microelectronics, University of Las Palmas de Gran Canaria

LPGC, Spain

Sep. 2021 - Nov. 2021

Teaching Experience in UASLP

Course “electricity and magnetism” in Bachelor’s Degree

S.L.P., Mexico

Aug. 2020 - Jan. 2021

Social Service

Assistant in the Laboratory of Basic Biomedical Engineering

S.L.P., Mexico

Aug. 2018 - Feb. 2019

Professional Practices

Assistant in the Laboratory of Basic Biomedical Engineering

S.L.P., Mexico

Oct. 2016 - Feb. 2017

Social Service

Service engineer in TECNHOSS Clinical Hospital Engineering

S.L.P., Mexico

Mar. 2016 - Sep. 2016

Awards and Honors

Oct. 2024	Honorable mention: Outstanding Reviewer, 27th International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI).	MA, MAR
Sep. 2023	Acknowledgment: Graduated with Honors in PhD in Engineering Sciences, Autonomous University of San Luis Potosí.	MX-SLP
Oct. 2022	Contest: First Prize in “Dr. Fernando Prieto Hernández” postgraduate student competition held during the National Congress of Biomedical Engineering	MX-JAL
Sep. - Nov 2021	Mobility Scholarship: “CUMex – AUIP 2021”	ES-GC
Oct. 2020	Contest: Second Prize in “Ing. Teófila Cadena Alfaro” graduate student competition held during the National Congress of Biomedical Engineering	Virtual Conference
Nov. 2017	Contest: First Prize in “InnovaHealth Challenge” bachelor student competition held during the National Congress of Biomedical Engineering	MX-NLE

Volunteer Services

28th International Conference on Medical Image Computing and Computer Assisted Intervention Society.

Web of Conference [\[Website\]](#)

Peer Reviewer

Sep 2025

27th International Conference on Medical Image Computing and Computer Assisted Intervention Society.

Web of Conference [\[Website\]](#)

Peer Reviewer

Oct 2024

Euromicro Conference on Digital System Design 2024

Web of Conference [\[Website\]](#)

Committee Member of the Special Session “Hyperspectral Imaging Applications, Algorithms and Architectures (HIAAAA)”.

Aug 2024

Euromicro Conference on Digital System Design 2023

Web of Conference [\[Website\]](#)

Committee Member of the Special Session “Hyperspectral Imaging Applications, Algorithms and Architectures (HIAAAA)”.

Sep 2023

IEEE Latin America Transactions

Web of Science [\[Website\]](#)

Peer Reviewer

Jan. 2023 - Present

IEEE Access

Web of Science [\[Website\]](#)

Peer Reviewer

Jan. 2023 - Present

45th National Congress of Biomedical Engineering

Web of Conference [\[Website\]](#)

Peer Reviewer

Nov 2022

Euromicro Conference on Digital System Design 2022

Web of Conference [\[Website\]](#)

Committee Member of the Special Session “Hyperspectral Imaging Applications, Algorithms and Architectures (HIAAAA)”.

Aug 2022

International Academic Conferences

Presentation / Attendance

- International Workshop on Graphs in Biomedical Image Analysis (GRAIL 2024), at MICCAI 2024 (Oct 2024), Marrakesh, Morocco.
- IEEE 20th International Symposium on Biomedical Imaging (ISBI) (Apr 2023), Cartagena, Colombia.
- 25th Euromicro Conference on Digital System Design (DSD) (Aug 2022), Maspalomas, Spain.
- 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (Nov 2021), Guadalajara, Mexico.